Homework Sets

15–20 minutes a week: Core, Challenge and Creative

Tip: How homework works

Do the **Core** task first (about 10 minutes). **Challenge** and **Creative** are optional extras for when you want more. Finish with the self-check list. If you get stuck for more than 5 minutes, write down what you tried and bring the question next week: that is a great result too.

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Week 1
Robots Need Instructions
Level 1: Introduction

15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Walk and Talk (Scratch, about 10 minutes)

Make a sprite walk across the stage and talk, in the right order. (No Mac at home today? Do the Challenge on paper instead and build this next week.)

1. Open the Scratch app. Click **Choose a Backdrop** (bottom right) and pick any place.
2. Build this script for the cat (or any sprite you choose). **glide** is in **Motion**.
3. Click the green flag. The sprite should start on the left, talk, glide to the middle, talk, glide to the right and say bye.
4. Save it: **File** → **Save to your computer**, name it `walk-and-talk`.



Use your own nickname and your own words in the speech bubbles.

★★ Challenge (want more?): Be the programmer for your family robot

Write an algorithm for brushing your teeth in exactly 6 steps. Then ask someone at home to be the robot and follow your steps **exactly**, without guessing. Did they do it right? If not, fix your steps.

What did you have to fix?

Creative (make it yours): A three-scene story

Add a “switch backdrop to …” block (in **Looks**) between the glides so the sprite walks into a new place each time. Choose a backdrop for each part of your story.

Self-check: tick each box when it is true

- ☐ My script starts with “when green flag clicked”.
- ☐ My sprite always starts on the left, because of the “go to x: -180 y: -100” block.
- ☐ The steps happen in the order I planned.
- ☐ I used my nickname, not my real name.
- ☐ I saved my project.

Week 2
Loop-de-Loop
Level 1: Introduction

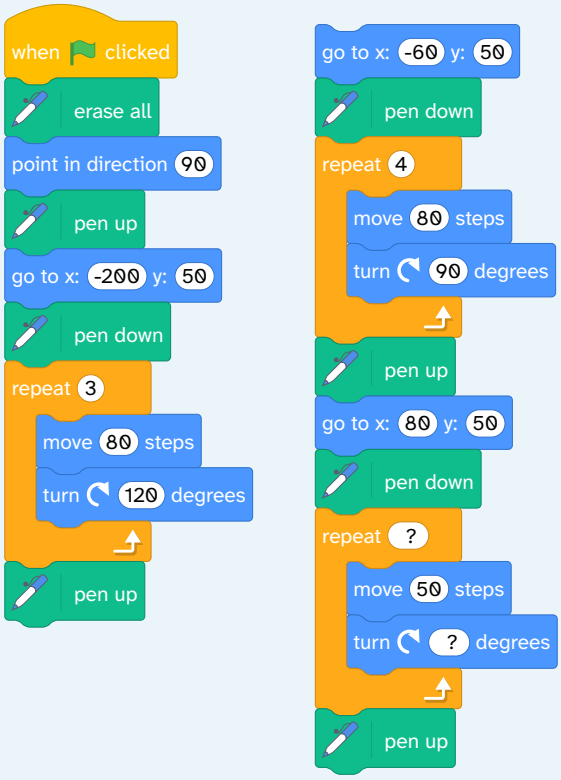
15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Shape Gallery (Scratch, about 10 minutes)

Draw a triangle, a square and a hexagon in a row, like pictures in a gallery.

1. Open Scratch. Click **Add Extension** (bottom left) and choose **Pen**. Set the cat's **Size** to 40.
2. Build the script. "pen up" before each "go to" stops the cat drawing a line between the shapes.
3. Fill in the two missing numbers for the hexagon (use the rule **turn = 360 ÷ sides**).
4. Click the flag. Save as `shape-gallery`.



(The right-hand stack snaps under the left-hand one: it is one long script, split in two to fit the page.)

My hexagon numbers: repeat _____ turn _____

★★ Challenge (want more?): The circle trick

Draw a shape with “repeat 36”, “move 10 steps”, “turn 10 degrees”. What does it look like? Why do you think it looks like that? (Hint: 36 × 10 = ?)

What turn would you need for a shape with 10 sides? _____

Creative (make it yours): Your own loop art

Make a picture with loops. Ideas: put “change pen color by 10” inside a loop; put “change pen size by 1” inside a loop; repeat a square 12 times and turn 30 degrees after each one. Draw a tiny sketch of your favorite result here:

Self-check: tick each box when it is true

- ☐ I used “repeat” instead of copying the same blocks again and again.
- ☐ My move and turn blocks are **inside** the repeat.
- ☐ Every shape closes (it ends where it started).
- ☐ I can explain the rule: turn = 360 ÷ number of sides.
- ☐ I did the 20-20-20 eye break when I used the screen.

Week 3
When This Happens...
Level 1: Introduction

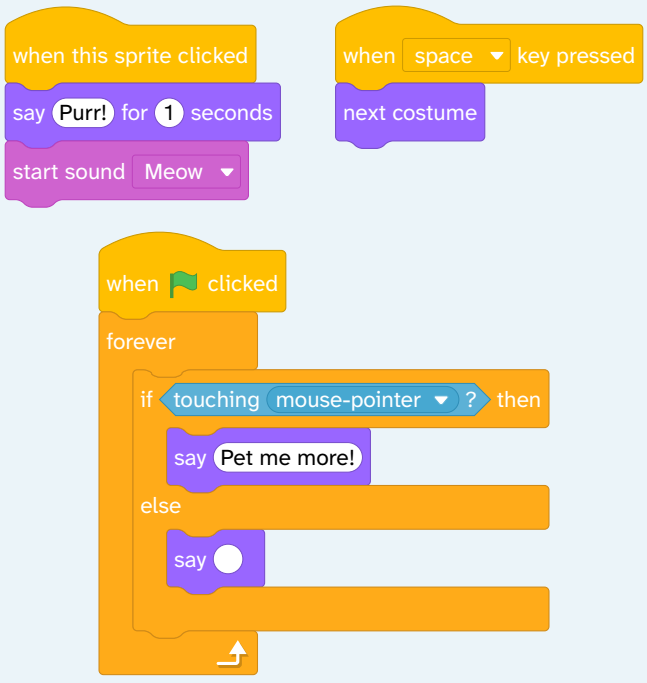
15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Pet Reactions (Scratch, about 10 minutes)

Make a virtual pet that reacts to you in three different ways.

1. Open Scratch. Delete the cat or keep it as your pet. Or click **Choose a Sprite** and pick any animal.
2. Build the three scripts below. Each one has a different doorbell (event).
3. Click the flag. Then: click the pet, press `space`, and move the mouse over the pet.
4. Save as `pet-reactions`.



(The sound block may have a different name for your animal: just pick any sound from the dropdown.)

★★ Challenge (want more?): Find the bug

Kai wants the pet to say “Wake up!” whenever the mouse touches it. Kai’s script is: “when green flag clicked” → “if touching mouse-pointer? then say Wake up!”. It never says “Wake up!”. Why not? How would you fix it?

Creative (make it yours): Two stars and a wish at home

Show your pet to someone at home. Ask them for two stars (two things they like) and a wish (one idea). Write them here, then try to make the wish come true!

★ _____

★ _____

Wish: _____

Self-check: tick each box when it is true

- ☐ My pet reacts to a click, a key press and the mouse.
- ☐ My “if” is inside a “forever”, so it keeps checking.
- ☐ I can say which block is the event (doorbell) in each script.
- ☐ When I gave or got feedback, it was kind and specific.

Week 4
Keeping Score
Level 1: Introduction

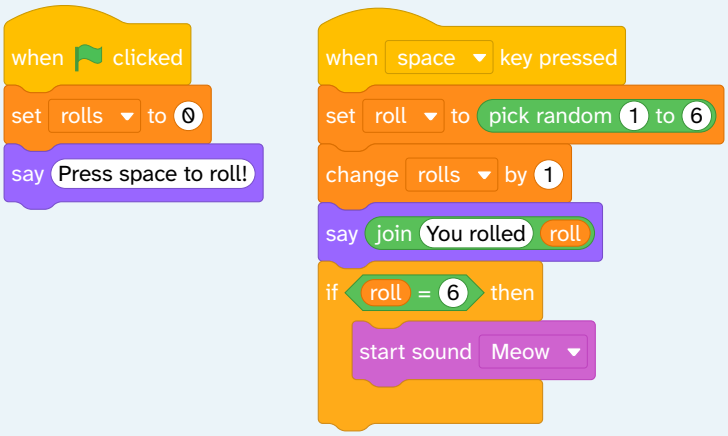
15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Magic Dice (Scratch, about 10 minutes)

Make a dice roller that remembers how many times you rolled.

1. Open Scratch. Make two variables (**Variables** → **Make a Variable**): `roll` and `rolls`.
2. Build the two scripts. "join" and "pick random" are in **Operators**. Type a space after "You rolled" so the words don't squash together.
3. Click the flag, then press `space` to roll.
4. Save as `magic-dice`.



Roll 10 times. Write your numbers: _____

Could you predict the next number? _____ What does `rolls` show now? _____

★★ Challenge (want more?): Doubles!

Add a second variable `roll2` and set it to "pick random 1 to 6" too. Make the sprite say "Doubles!" when `roll` = `roll2`. Roll until you get doubles. How many rolls did it take? _____

Bug check: Zoe put "set rolls to 0" inside the "when space key pressed" script. What will `rolls` always show? Why?

_____

Creative (make it yours): Real dice faces

Click the **Costumes** tab and paint 6 costumes, one for each dice face, with 1 to 6 dots (dots, not colors, show the number). Put them in order. Then add "switch costume to (roll)" to your space-key script so the dice shows the face you rolled. Credit yourself: add a comment "Dice faces drawn by" + your nickname.

Self-check: tick each box when it is true

- ☐ I made variables and chose "For all sprites".
- ☐ `rolls` starts at `0` every time I click the green flag.
- ☐ `rolls` goes up by exactly 1 for each roll.
- ☐ I can explain the difference between "set" and "change".

Week 5
Messages and My Blocks
Level 1: Introduction

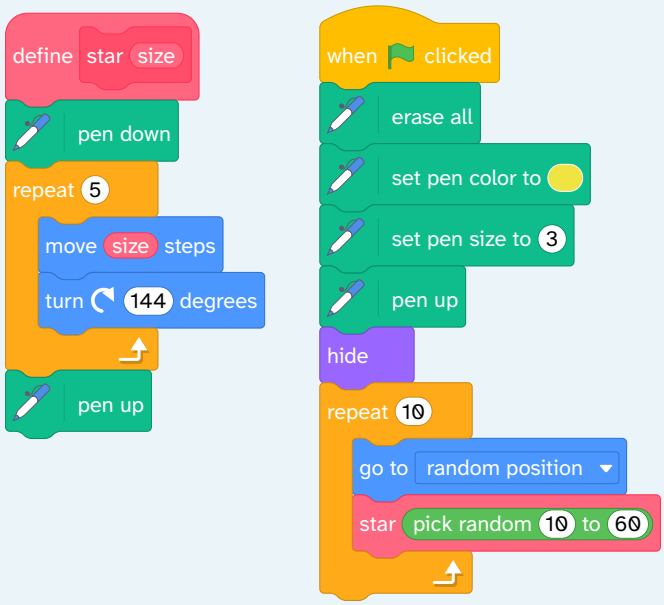
15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Starry Sky (Scratch, about 10 minutes)

Make your own “star” block and fill the night sky.

1. Open Scratch and add the **Pen** extension. Pick a dark backdrop (for example **Stars**, or paint the backdrop dark blue).
2. **My Blocks** → **Make a Block**: name it `star`, add an input `size`, click **OK**.
3. Build both scripts. Drag `size` out of the define hat into the move block. For the pen color, click the color circle and set **Color 16, Saturation 72, Brightness 94** (yellow from the color-safe paint box).
4. Click the flag. Save as `starry-sky`.



Why does the star block turn 144 degrees and not 72? (Hint: count how many times the pen goes all the way round.)

★★ Challenge (want more?): Add a moon (decomposition)

Make a second My Block called `moon` with no input. It should go to the top-left corner and draw a big circle: repeat 36, move 8, turn 10. Use it in your green-flag script before the stars. Now your main script reads like a plan: erase all, moon, stars. Write your main script as a plan, one line per block:

Creative (make it yours): Good night message

Add a **Button** sprite (any sprite) that says “Click me for night time”. When it is clicked, it broadcasts a new message `night falls`. Give the cat a “when I receive night falls” script that says “Good night!”. Check that both blocks use exactly the same message.

Self-check: tick each box when it is true

- ☐ My `star` block has an input called `size`, and I used it inside.
- ☐ Every click on the flag draws 10 stars of different sizes in different places.
- ☐ My stars can be seen clearly on the dark sky (I used a color from the paint box).
- ☐ (Creative) My broadcast and my “when I receive” use the same message.

Week 6 Level 1: Introduction

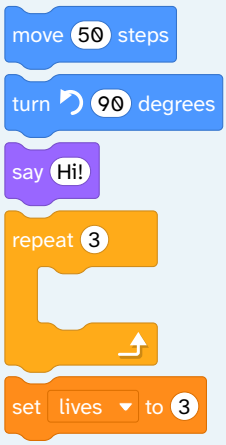
Review Studio and Checkpoint

15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Remix Studio (about 10 minutes)

Part 1: Blocks to words (paper, 4 minutes). Use your card sort to write the Python line for each block. Guess if you are not sure: guessing is good practice!



_____

_____

_____

_____

_____

Part 2: Level up a project (Scratch, 6 minutes). Open one of your projects from weeks 1–5 (**File** → **Load from your computer**). Add **one** improvement from this menu, test it, and save it with a new name ending in `-v2`: a sound effect • a new backdrop • a score or lives variable • a key that does something • a My Block • a message between two sprites.

My project: _____ What I added: _____

★★ Challenge (want more?): A bug for a friend

Build a tiny project (8 blocks or fewer) with **one** bug hidden inside. On a card, write what it **should** do, and draw or list its blocks. Bring the card to the next session: a classmate will try to find your bug!

It should:

The bug (keep this secret, fold it over):

Creative (make it yours): Remix with a new theme

Make a remix of Apple Catch with a new theme, for example catching stars in space, or fish in the sea. Change the sprites, the backdrop and the sounds (from the Scratch library or drawn by you). Add a credits comment: “Remix of Apple Catch by” + your nickname, + “sprites from the Scratch library”.

Self-check: tick each box when it is true

- ☐ I wrote a Python line for every block in Part 1.
- ☐ I added one improvement to an old project and it works.
- ☐ I saved the new version with `-v2` at the end of the name.
- ☐ My remix (if I made one) has a credits comment.

Week 7
Triangle Time
Level 1: Introduction

15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Triangle Time (about 10 minutes)

In Thonny, open `week07/hw-triangle/starter.py`. The turtle already draws **one** side.

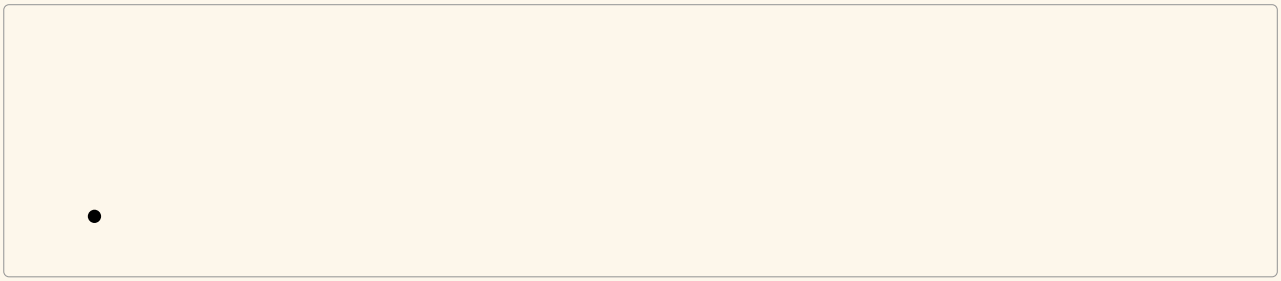
1. Add the code for the other two sides. Each side is `t.forward(120)` then `t.left(120)`.
2. Why 120? The turtle must turn all the way round (360 degrees) in 3 corners: $3 \times 120 = 360$.
3. Press Run. Then open `check_my_work.py` and press Run.

No computer at home? Write the whole program here and type it in next week:

★★ Challenge (want more?): Be the turtle

Don't run it yet! Read this program and draw what the turtle will draw in the box. The turtle starts at the dot, facing right. Then type it in and check.

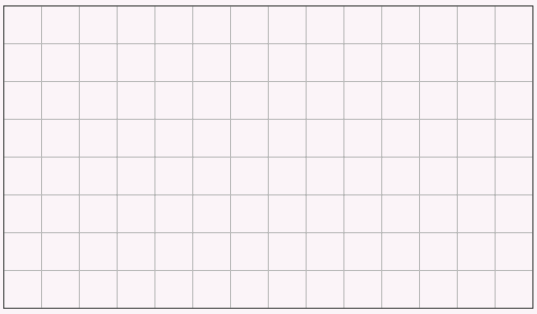
```
t.forward(100)
t.left(90)
t.forward(50)
t.left(90)
t.forward(100)
```



What letter or shape does it look like? _____

Creative (make it yours): My initial

Plan the first letter of your nickname with straight lines only (E, F, H, L, T, Z work well). Draw it on the grid. Write the steps next to it, like `F 100, L 90, F 50`. Then try it in Thonny in a new file saved as `my-python/initial.py`.



My steps:

Self-check: tick each box when it is true

- ☐ My triangle has 3 equal sides and is closed.
- ☐ I ran `check_my_work.py` (or wrote my code on paper).
- ☐ I saved my file with a sensible name in the right folder.
- ☐ If I met a red error, I read the line number and the last line.

Week 8
Loop Fixer
Level 1: Introduction

15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Loop Fixer (about 10 minutes)

Open `week08/hw-loop-fixer/starter.py`. It should draw a blue square and an orange star, but it has **2 loop bugs**.

1. Press Run and watch carefully. Count the sides of the square. How many times does its loop repeat?
2. The star is a straight line! Look at the spaces at the start of each line. Which line is **not** inside the loop?
3. Fix one bug, press Run, then fix the other. Finish with `check_my_work.py`.

Bug 1 was on line _____. I changed _____ to _____.

Bug 2 was on line _____. I fixed it by _____.

No computer at home? Circle the two bugs in the program below and write the fixes next to them.

```
for i in range(3):
    t.forward(100)
    t.left(90)

for i in range(5):
    t.forward(150)
t.right(144)
```

★★ Challenge (want more?): Count the taps

Without running it: how many times does this program print `tap`? _____

```
for i in range(3):
    for j in range(2):
        print("tap")
```

Then make a 7-pointed star: 7 lines of 150, turning `t.right(360 * 3 / 7)` each time (about 154 degrees).

Creative (make it yours): Spiral lab

Make a new file `my-python/spiral-lab.py` with this loop (plus `import turtle`, `t = turtle.Turtle()` and `turtle.done()`). Change the turn number and record what you see.

```
for i in range(60):
    t.forward(i * 3)
    t.left(91)
```

Turn	What it looked like
91	
121	
my choice:	

Self-check: tick each box when it is true

- ☐ My square has 4 sides and my star has 5 points.
- ☐ I can explain why the spaces at the start of a line matter.
- ☐ I know that `range(4)` makes a loop run 4 times.
- ☐ If I got stuck, I used the help sentence: “I expected … but … happened.”

Week 9

Mini Math Quiz

Level 1: Introduction

15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Mini Math Quiz (about 10 minutes)

Open `week09/hw-math-quiz/starter.py`. No turtle this time: everything happens in the Shell.

1. Press Run and answer question 1 in the Shell. Read how it works: `input`, then `int`, then `if/else`, and `score = score + 1` when the answer is right.
2. Add question 2 the same way: "What is 9 + 15?" (the answer is 24).
3. At the end, print the score: `print("You scored", score, "out of 2")`.
4. Test it twice: once with right answers, once with wrong ones. Then run `check_my_work.py`.

No computer at home? Write question 2 here (copy the pattern of question 1):

★★ Challenge (want more?): Perfect score

Add a third question of your own. Change "out of 2" to "out of 3". Then, at the very end, add an `if` that prints a special message when `score == 3`. Remember: two equals signs ask a question!

Creative (make it yours): Choose your path

Write a tiny adventure: the program asks "You find a door and a tunnel. Which do you choose?" and uses `if/elif/else` to print a different ending for `door`, `tunnel` and anything else. Plan it here first:

Remember the rule: our programs ask about **choices**, never about who you are (no names, addresses, schools or passwords).

Self-check: tick each box when it is true

- ☐ My quiz asks 2 questions and says `Correct!` for right answers.
- ☐ The last line shows my score.
- ☐ I know why we need `int(...)` around `input(...)` for numbers.
- ☐ I know the difference between = (put in the box) and == (are they the same?).

Week 10 Level 1: Introduction

My Art Show Plan

15–20 minutes. This week's homework is your project plan (Milestone M1).

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Finish your plan sheet (about 10 minutes)

Use the **plan sheet** in your Project Brief (page 4). Make sure it has:

- ☐ a **sketch** of your picture, with the parts labeled;
- ☐ your **question(s)**: 1 or 2 questions about choices (a number, a color, day or night);
- ☐ your **function**: its name and its input slots, like `fish(x, y, color)`;
- ☐ a few words for each ingredient: my loop repeats..., my `if` decides..., my random surprise is..., my variable remembers...

Bring the plan sheet next week. You will build from it!

★★ Challenge (want more?): Write your function on paper

Write the recipe for your function. Start with the line below, then add the steps (penup, goto, pendown, then the drawing). If you have a computer, try it in your project file.

`def` _____`(x, y,` _____`):`

Creative (make it yours): Colors and a title

Choose 3 colors from the color-blind-safe palette for your picture and give your artwork a title.

orange #E69F00	blue #0072B2
sky blue #56B4E9	vermillion #D55E00
green #009E73	purple #CC79A7
yellow #F0E442	black #000000

My 3 colors: _____

The title of my artwork: _____

Self-check: tick each box when it is true

- ☐ My sketch shows what my program will draw.
- ☐ My questions are about choices, not about who someone is.
- ☐ My function has a name that says what it draws.
- ☐ I know where each of the six ingredients will go.

Week 11 Level 1: Introduction

Get Ready for the Showcase

15–20 minutes. This week's homework is project work: finish and practice.

Name (nickname is fine): _____ Date: _____

⭐ Core (everyone): Finish and practice (about 10 minutes)

1. If your project is missing an ingredient, add it now (or write on paper how you will add it next week). Which ingredients still need work?

   _____

2. Practice your **2-minute demo** out loud for someone at home. They are the visitor: let them answer your program's question (or pretend, using your sketch). Use the script in your workbook (chapter 12).
3. Ask your visitor for **two stars and a wish**:

   ⭐ _____

   ⭐ _____

   🪄 _____

⭐⭐ Challenge (want more?): Be ready for strange answers

Visitors sometimes type surprising things. Make your program cope:

- a number that is too big? Use an `if` to make it smaller and print a friendly message;
- `Night` instead of `night`? Add `.lower()` after `input(...)`.

🎨 Creative (make it yours): Make your stand card

Fold a piece of paper in half to make a card that stands up. Write your project's title, your **nickname** as the artist, and one sentence about your artwork. Decorate it! (No real full names, please: nicknames only.)

✅ Self-check: tick each box when it is true

- ☐ My program runs from start to end without red errors.
- ☐ All six ingredients are checked in the comment list at the top of my file.
- ☐ I practiced my 2-minute demo out loud.

☐ I can point to my function, my loop and my `if`.

Week 12 Level 1: Introduction

Look Back, Keep Coding

15–20 minutes for Core. The holiday menu is for whenever you feel like coding.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): My Level 1 reflection (about 10 minutes)

The thing I am proudest of in Level 1:

Something that was hard at first but is easier now:

Something I want to learn or make in Level 2:

★★ Challenge (want more?): Holiday "keep coding" menu: pick 2

No internet or account needed. Save each one in `Documents/my-python`.

1. **Magic spiral**: `for i in range(60):` with `t.forward(i * 3)` and `t.left(91)`. Try other turns.
2. **Kaleidoscope**: 36 squares, turning `t.left(10)` after each, with `random.choice(PALETTE)` colors.
3. **Dice roller**: `roll = random.randint(1, 6)`; print it; if it is 6, print "Roll again!".
4. **City skyline**: a `def building(height, color):` that draws a filled rectangle, called 12 times with random heights.
5. **Times-table quiz**: your Mini Math Quiz with 5 questions and a message for each score.
6. **Remix your art show**: add a new function, or a second question.
7. **Unplugged**: teach someone at home the "parent-robot" game from week 1 — you give the instructions!

I chose menu numbers _____ and _____. My favorite was _____ because _____.

Creative (make it yours): Invent a challenge

Invent a coding challenge for a friend or someone in your family. What should the program draw or do? Which ingredients does it need?

Self-check: tick each box when it is true

- ☐ I wrote my Level 1 reflection.
- ☐ I picked at least one menu challenge to try over the holiday.
- ☐ My programs are saved on my computer, with sensible names.
- ☐ I am a Level 1 coder!